ACTION PLAN V4 · SUPERSEDES V3

# RAIN: Routes You Can Trust

**Risk-Aware Indoor Navigation**

Phone-only indoor navigation for buildings with almost no Wi-Fi and unreliable signs — where the route is planned around how well the system can guide it, not just how short it is

| | |
|---|---|
| **TEAM** | Khwaja Mohammed Ifzal · Mohammed Saheel · Syed Musid · Talha Khan |
| **DEPT** | CSE (AI & ML), Mangalore Institute of Technology & Engineering · GUIDE Dr. Farha Haneef |
| **SCOPE** | Main Block, single floor, approx. 20 rooms · plus a multi-college Wi-Fi snapshot |
| **STATUS** | Phase II, pre-implementation |
| **PAPER TITLE** | **Risk-Aware Indoor Navigation for Infrastructure-Poor Buildings: Planning Routes a Smartphone Can Reliably Guide** |
| **REVIEW TITLE** | Indoor Navigation Beyond Dense Infrastructure: A Review of Localisation, Sign Reading and Route Planning in Resource-Constrained Buildings |
| **DELIVERABLES** | Literature review paper · Technical paper · Working demo · Public dataset |
| **SUPERSEDES** | Action Plan v3. Team unchanged. **Project renamed from SHINS 2.0 to RAIN** to reflect the navigation-side focus; confirm the change with the guide and department records. |

Action Plan v3 fixed the **order** of the work. v4 changes two deeper things. First, the regime: our own college, like many Indian colleges, has only **one or two access points**, so the system must work with Wi-Fi as an optional extra — including with none at all. Second, the headline novelty moves from *localisation* (where am I?) to *navigation* (which way should I send the user, and when should I ask them to check?). Every sound decision from v3 is kept; the deep learning and machine learning now each have a clear job.

**THE SHIFT IN ONE SENTENCE**

**Don't send users along the shortest path and hope positioning keeps up. Plan the route the system can reliably guide, and ask the user to check only where it matters.**

**HONEST RISK ASSESSMENT**

The biggest risks are no longer weak novelty. They are: (1) the navigation trial with real participants — it is the proof of the headline claim, so it needs guide approval and consent planning early; (2) claims that outrun the evidence — every “no study has…” must survive a documented search; and (3) building impressive models before the basic app can read a sign and draw a route.

## 01 The problem and what we are solving

A new student, visitor or parent walks into an Indian college building and cannot find Room 204. GPS does not work indoors, and there is often no one to ask. Existing indoor navigation assumes things these buildings do not have:

- **Many Wi-Fi access points.** Research systems need dozens or hundreds. Many colleges have one or two, and most visible networks are students' phone hotspots that disappear when class ends.
- **Bluetooth beacons or installed hardware**, which colleges will not pay for.
- **Official digital floor plans**, which commercial indoor maps depend on and colleges rarely provide.
- **Clear, readable signs.** Real nameplates are faded, handwritten, covered by notices, affected by glare, or missing.

So existing approaches fail twice. The phone cannot reliably tell *where* the user is, and even when it can, the shortest route often passes confusing junctions and unreadable signs — exactly where people get lost.

### Research question

> **How can a phone reliably guide someone to the right room in a building with 0–2 access points and unreliable signage, using no extra hardware and minimal user effort?**

### Four sub-problems

| Question | What we build | AI / ML | Novelty weight |
|---|---|---|---|
| **Which way should I go?** | Risk-aware route planning + off-route detection and re-planning | ML risk model + classifier | Headline |
| **When should I check?** | Uncertainty-aware scan prompting, sharing the same risk model | ML policy | Strong |
| **Where am I?** | Room recognition from sign text and doorway appearance, fused | Deep learning | Moderate |
| **Where am I between signs?** | Step counting and heading, snapped to the corridor graph | Classical | Plumbing |

Supporting all four: a **measurement study** of the few-AP, hotspot-heavy regime across several colleges, and a **public benchmark dataset** of degraded signage, labelled Wi-Fi scans, motion traces and navigation-trial logs.

### Problem statement for the paper

> **USE THIS WORDING AS THE STARTING DRAFT**
>
> *Indoor navigation systems typically assume dense Wi-Fi infrastructure, installed beacons or venue-supplied floor plans, and plan routes by shortest distance. Many academic buildings in India violate all of these assumptions: they have one or two access points, radio environments dominated by transient personal hotspots, and degraded or missing room signage. Under these conditions, radio-based localisation cannot identify rooms, and shortest-path routes pass through locations where users are likely to go wrong. This work addresses the problem of reliable room-level navigation in such infrastructure-poor buildings using only a smartphone, by recognising rooms visually, tracking movement between recognitions, and planning routes that trade distance against the learned risk of navigation error.*

## 02 What changed from v3

| Area | Change in v4 |
|---|---|
| **Regime** | Assumes **0–2 access points**, not three. Wi-Fi becomes an optional input; the system must work with none. |
| **Headline novelty** | **NEW — Risk-aware route planning.** Routes minimise distance plus the learned risk of the user going wrong. Moves the paper onto the navigation side of the field. |
| **Joint decision** | **NEW — One risk model drives two decisions:** avoid a risky junction, or keep the route and prompt a scan before it. |
| **Off-route handling** | **NEW — Off-route detection and re-planning** from step and heading patterns. |
| **Between-sign tracking** | **NEW — Motion tracking** (pedestrian dead reckoning) snapped to the corridor graph replaces Wi-Fi as the between-sign position source. |
| **Room recognition** | **NEW — Fused doorway recognition.** A pretrained image embedder recognises the doorway's appearance alongside OCR, covering missing and obscured signs. |
| **Prompting policy** | Features change from Wi-Fi-driven to **drift since last fix, junction distance, recognition confidence**, with Wi-Fi as optional features. Adds a with/without-Wi-Fi ablation. |
| **AP-scarcity curve** | Extended down to **2, 1 and 0 APs**. |
| **Evidence for the regime** | **Multi-college AP snapshot promoted from optional to core.** “Many colleges have few APs” must be measured, not asserted. |
| **Hotspot landmarks** | Promoted from side experiment to **central**: with 1–2 APs, hotspots are nearly everything visible. |

| Area | Change in v4 |
|---|---|
| **Dataset** | Adds **motion traces and navigation-trial logs**. |
| **Evaluation** | Adds a **participant navigation trial**: shortest vs simplest vs risk-aware routes. |
| **Numbers to re-check** | The 86% hotspot figure and the “3 infrastructure APs” lab count must be re-measured and reconciled with the 1–2 AP reality (see Section 05). |

# 03 Novelty claims, ranked and stress-tested

Ranked by strength. Each claim is written in the narrow form it can defend. The order also gives the paper its structure: **measure the regime → recognise rooms → plan guidable routes → decide when to check.**

| # | Claim | Strength | Closest prior art to read |
|---|---|---|---|
| **1** | Risk-aware routing with learned risk, joined to scan prompting | Strong, navigation-side | Simplest path (Duckham & Kulik); least-risk path (Grum; Vanclooster et al.) |
| **2** | Uncertainty-aware scan prompting as a cost-sensitive decision | Strong | Conformal indoor positioning; active querying in robotics / HCI |
| **3** | 0–2 AP regime measurement + scarcity curve to zero | Solid, data-driven | XJTLU dynamic database; EdgeLoc; Locaris; single-AP datasets |
| **4** | Fused text + appearance recognition on degraded Indian signage | Moderate | OCRAPOSE; TextInPlace; NetVLAD-style place recognition |
| **5** | Public benchmark dataset | Moderate, high citation value | UJIIndoorLoc; SODIndoorLoc; Mendoza-Silva long-term set |
| **6** | Session-scoped hotspot landmarks | Exploratory, cannot fail | Hotspot-filtering literature |

### Claim 1 — Risk-aware routing. Headline.

**Known:** trading distance for fewer navigation errors is established — “simplest path” and “least-risk path” routing use hand-set rules about turns and junction complexity. **Surviving claim:** risk weights *learned from measured data* (sign readability, motion drift, junction confusion), shared with the scan-prompting policy so the system chooses between *avoiding* a risky place and *checking* before it, in a phone-only system with 0–2 APs. **Never say** “we invented safe routing.”

### Claim 2 — Uncertainty-aware prompting. Holds.

Confidence estimation for localisation exists, and “when should a system ask the user” is studied in robotics and HCI. Casting sign-scan prompting in a phone-only navigator as a cost-sensitive decision driven by drift and route risk was not found published. The with/without-Wi-Fi ablation adds a clean question: *do one or two access points add anything at all?* Either answer is a result.

### Claim 3 — The regime. Holds, if it is measured.

Sparse-AP localisation and hotspot contamination are both known. What is not documented is a building-scale, multi-college characterisation of buildings with 0–2 APs, and an accuracy curve pushed all the way to zero APs. Single-AP fingerprinting studies exist — cite them, do not claim a first.

### Claim 4 — Fused recognition. Narrow it.

Reading room numbers for positioning (OCRAPOSE) and combining text with place recognition (TextInPlace) are prior art. Surviving claim: a **condition-stratified study** of text-only, appearance-only and fused recognition on degraded Indian college signage, on a phone, with few-shot enrolment, tied to navigation outcome. Read TextInPlace closely before writing this.

### Claims 5 and 6

The dataset is new because nothing pairs degraded Indian signage with hotspot-labelled RF, motion traces and trial logs. Hotspot landmarks invert the usual “filter them out” assumption; a negative result strengthens Claim 3.

**WHAT WE ARE EXPLICITLY NOT CLAIMING — SAY THIS BEFORE ANYONE ASKS**

Not the first to use a camera for fingerprint labelling (Heimdall). Not the first to auto-collect fingerprints from wall markers (QRLoc). Not the first to grow a map from ordinary use (Redpin). Not the first to use semantic text anchors (LLM-Loc). Not the first to read room numbers for positioning (OCRAPOSE, TextInPlace). Not the first to reset dead reckoning at landmarks (e.g. UnLoc). Not the first to route for simplicity or low risk (Duckham & Kulik; Grum). Not the first to fingerprint with a single AP. Not claiming to beat commercial accuracy.

# 04 System architecture

The camera and motion sensors locate the user. A single shared risk model then drives both navigation decisions. Wi-Fi is never trusted to name a room; with 1–2 APs it is at most an extra feature.

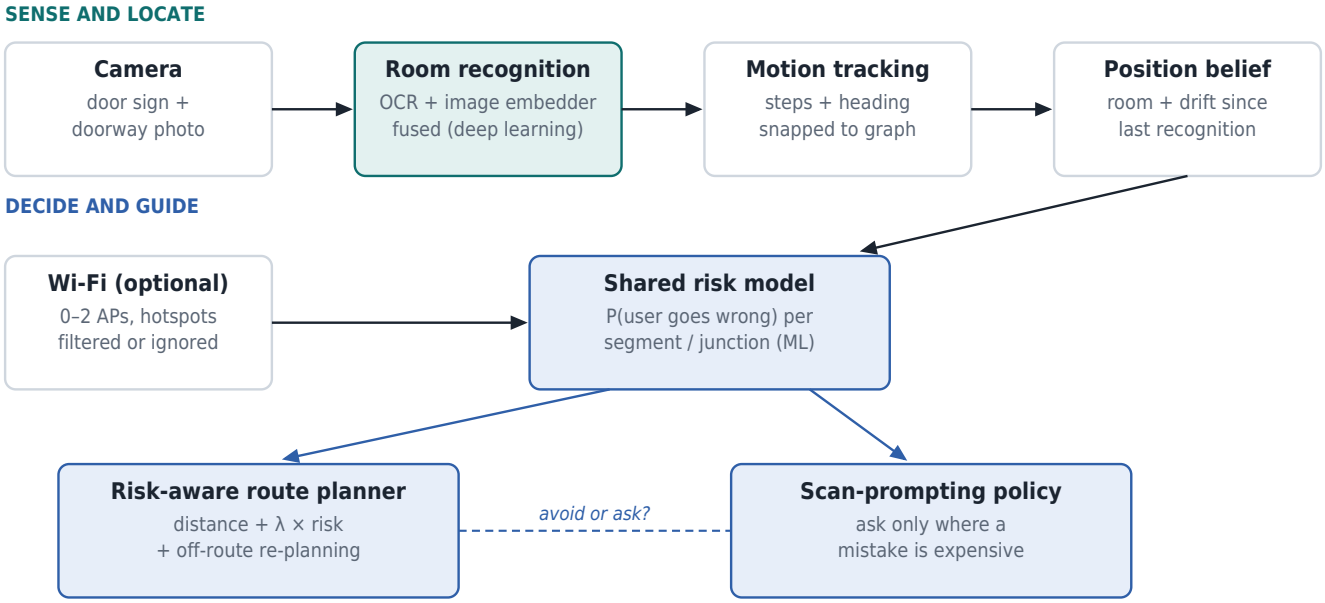



*Figure 1. RAIN pipeline. Blue components are the navigation-side contributions.*

| Layer | Component | How it works | Runs |
|---|---|---|---|
| **Map** | Topological floor graph | rooms.csv (nodes) and edges.csv (corridor segments, junctions, lengths) from the floor walk. SQLite on device. | On-device |
| **Locate** | Fused room recognition | ML Kit OCR + fuzzy match to known room names, combined with an image-embedding match against 3–5 enrolment photos per doorway. | On-device |
| **Locate** | Motion tracking | Step detection and heading from accelerometer and gyroscope; position advanced along the current edge and snapped to the graph; reset at each recognition. | On-device |
| **Decide** | Shared risk model | Gradient boosting predicts P(user goes wrong) per segment and junction from sign readability, drift, junction complexity and optional Wi-Fi features. | Trained offline, runs on-device |
| **Guide** | Risk-aware planner | Dijkstra over edge cost = distance + λ × risk. λ tunable; λ = 0 gives shortest path. | On-device |
| **Guide** | Scan-prompting policy | Prompts when expected cost of an error ahead exceeds the cost of interrupting. | On-device |
| **Guide** | Off-route detector | Compares observed turn and step pattern with the expected route; triggers re-plan or scan prompt. | On-device |

# 05 The contributions in detail

## C1 · Risk-aware route planning — the headline

Normal routing picks the shortest path. In this building, the shortest path may pass faded signs, a long corridor with no landmarks (so motion error builds up), and a four-way junction with identical doors. A slightly longer route might get the user there first time.

**EDGE COST**

$$\text{cost}(e) = \text{distance}(e) + \lambda \times \text{risk}(e)$$

risk(e) is the predicted probability that a user goes wrong on segment or junction e. λ controls how many extra metres the planner will spend to avoid one expected error.

**Risk features**, all from data already collected:

- Sign readability at doors along the segment — measured OCR and recognition success per door, by condition.
- Distance since the last recognisable landmark — a proxy for accumulated motion drift.
- Junction complexity — number of branches, turn angle, number of similar-looking doors nearby.
- Optional Wi-Fi features — whether any infrastructure AP is visible there.

**Training labels.** Seed labels come from pilot walks: team members (and later participants) walk routes with the app logging; every wrong turn, hesitation stop and off-route detection is tagged to the segment or junction where it happened. Start with gradient boosting or logistic regression — small tabular data suits them. If labels are too few early on, begin with a transparent hand-weighted score and replace it with the learned model once trial logs exist; report both.

**The joint decision.** For each risky place on a candidate route, the system compares two options: detour around it (cost: extra distance) or keep the route and prompt a scan just before it (cost: one interruption). It picks whichever is cheaper. This avoid-or-ask trade-off is the heart of Claim 1.

**Off-route detection and re-planning.** A lightweight classifier compares the user's turn events and step count with what the route expects. On a likely wrong turn it re-plans from the estimated position, or prompts a scan if the position is too uncertain. Wrong-way detection exists in pedestrian navigation, so this is a system component with a measurable result (detection delay), not a headline claim.

**Closing the loop (extended).** Update segment risk from wrong turns and off-route events logged in use, so the route map improves over time. Redpin-style “improves with use” is related prior art; keep the claim small.

**THREE THINGS THAT MAKE C1 CONVINCING**

**Beat the right baseline.** Beating shortest path is easy. Matching or beating *simplest path* (fewest turns) is what shows learned risk adds something.

**Plot the trade-off curve.** Sweep λ and plot extra distance walked against arrival success. One figure shows what reliability costs.

**Report either outcome.** “5% longer routes, far fewer wrong turns” and “the shortest path was already safe in this building” are both publishable.

## C2 · Uncertainty-aware scan prompting

A small model estimates the probability that the current position belief is wrong, from: distance walked since last recognition, number of turns since then, recognition confidence at the last fix, distance to the next risky place on the route, and optional Wi-Fi features. The app prompts only when the expected cost of being wrong exceeds the cost of interrupting.

- **Baselines:** always scan at every door; fixed-interval scanning; never prompt.
- **Figure:** prompts per journey against wrong-room rate, for each policy.
- **Ablation:** policy with and without Wi-Fi features — do 1–2 APs help?

- **Model:** logistic regression or gradient boosting first; a small LSTM or temporal CNN as an honest ablation; conformal prediction on top for coverage guarantees.

## C3 · The 0–2 AP regime, measured

- **Multi-college AP snapshot (core).** A simple scan-logger app run for about ten minutes per floor by friends at 5–10 other colleges. Record visible BSSIDs, infrastructure versus hotspot split (locally-administered MAC bit), and time and date. This turns an anecdote into evidence and answers “only one building.”
- **Home-building churn.** Multi-session, multi-day captures per room; persistence of each BSSID across sessions.
- **AP-scarcity curve to zero.** Subsample public datasets from hundreds of APs down to 8, 6, 4, 3, 2 and 1, and plot room-level accuracy for k-NN, MLP, small CNN and autoencoder. At zero APs, Wi-Fi accuracy equals chance, so report the camera-plus-motion system on the same axis for comparison.
- **Throttling re-check.** Before reporting any selective-hiding observation, repeat throttled and unthrottled captures at different times and rule out Android returning cached results.

> **RECONCILE THE NUMBERS BEFORE THEY REACH A PAPER**
> v3 recorded **3 infrastructure APs** in the lab; the team now describes the college as having **one or two**. These may both be true: one physical access point often broadcasts several BSSIDs (2.4 GHz and 5 GHz bands, multiple SSIDs). Group BSSIDs by shared MAC prefix and physical location to count **physical APs** versus **BSSIDs**, and report both. Re-compute the 86% hotspot share on the same basis.

## C4 · Fused doorway recognition

If OCR cannot read a sign, a text-only system is blind — and two of the six sign conditions (missing, obscured) can never be read. A pretrained image embedder (e.g. MediaPipe Image Embedder with a MobileNet backbone) turns a doorway photo into a vector and matches it against 3–5 enrolment photos per room. No training is needed, and adding a room means taking photos, not retraining.

| Sign condition | What dominates the decision |
|---|---|
| **Printed clear** | OCR text |
| **Printed faded / handwritten / glare** | OCR and appearance vote, weighted by their confidences |
| **Obscured / missing** | Appearance match |

**Experiment:** text-only vs appearance-only vs fused, stratified by the six conditions, with top-1 accuracy and confident-wrong rate. **Known risk:** corridors of identical doors may confuse the embedder. Measure it; the graph and motion prior can break ties, and a low-confidence match feeds the prompting policy instead of risking a wrong turn.

## C5 · Public benchmark dataset

Signage images labelled by condition; doorway enrolment photos; RF scans with per-BSSID infrastructure/hotspot labels; multi-session churn; multi-college AP snapshots; motion traces; anonymised navigation-trial logs. Released with a datasheet covering collection method, devices and limitations.

> **PRIVACY — A PUBLICATION GATE, NOT A FORMALITY**
> Hotspot BSSIDs and SSIDs are personal data. Hash or truncate every BSSID and redact every SSID **at capture**, inside the logger, so no raw identifiers ever sit on a laptop. Photos must exclude faces or blur them. Trial logs carry participant codes, never names. Participants give written consent. State data handling explicitly in the paper.

## C6 · Session-scoped hotspot landmarks

Every paper treats hotspots as noise. Within one lecture hour a hotspot may stay put long enough to act as a short-lived landmark between sign reads — seeded at each recognition, expired on churn. Measure within-session versus across-day persistence. With only 1–2 APs, this is the only radio signal with any density, so the result matters more than in v3. A negative result reinforces C3.

# 06 Where the AI, ML and deep learning live

**THE RULE THAT DECIDES EVERYTHING**

**Deep learning where perception happens. Machine learning where decisions are made. Classical methods where data is too thin.** The camera reading a degraded sign is a perception problem. Choosing routes and prompts from tabular features is a small-data decision problem. Wi-Fi with 0–2 APs is not a learning problem at all — and we prove that with a controlled experiment instead of asserting it.

| Component | Type | Role | Where it runs |
|---|---|---|---|
| **Shared risk model** | ML (gradient boosting) | Predicts where users go wrong; feeds routing and prompting | Trained offline, on-device inference |
| **Risk-aware planner** | ML-weighted graph search | Chooses guidable routes | On-device |
| **Off-route detector** | ML classifier | Detects wrong turns early | On-device |
| **Scan-prompting policy** | ML (+ conformal, LSTM ablation) | Decides when to ask the user | On-device |
| **Fused doorway recognition** | Deep learning (pretrained CNN embedder) | Identifies the room when signs fail | On-device |
| **OCR model comparison** | Deep learning | Benchmarks which reader works on which signs | Offline, laptop |
| **AP-scarcity curve** | ML vs DL | Justifies keeping Wi-Fi classical | Offline, public data |
| **Fine-tuned on-device OCR** | Deep learning | Upgrade for handwritten and faded signs | Extended phase |
| **Neural OCR corrector** | Deep learning | Compared against weighted edit distance | Extended phase |

### OCR model comparison (offline)

Photograph all twenty nameplates in real corridor lighting, no flash. Run the same images through Google ML Kit (baseline; itself a CNN detector with a sequence decoder), a PP-OCR-class mobile model, and a heavier model such as TrOCR-Small (laptop-only accuracy ceiling). Report accuracy by sign condition, model size and latency. Ship ML Kit in the app regardless. Re-measure all sizes and latencies on your own phones.

### Fine-tuning and the neural corrector (extended)

With twenty nameplates, never train from scratch: generate synthetic images of your own room names with glare, blur, motion, skew, low light and occlusion matched to the six conditions, then fine-tune a pretrained recogniser. Converting it to TFLite or ONNX on a mid-range phone is fiddly — treat it as an upgrade to a working app, never a dependency. For the corrector, the vocabulary is closed (twenty names), so weighted edit distance is hard to beat; report honestly if it wins.

**CORRECTION CARRIED FROM V3**

Do not claim deep learning “never” beats k-NN at very few APs. Published evidence leans the other way at extreme scarcity, and no verified study gives a clean head-to-head at 1–3 APs. Present the curve as **your measured result**, and justify keeping k-NN on grounds of on-device incremental updates and tiny training data.

# 07 Evaluation plan

| # | Experiment | Data | Key metrics | Supports |
|---|---|---|---|---|
| **E1** | Multi-college AP snapshot | 5–10 colleges, scan logger | Physical APs, BSSIDs, hotspot share per floor | C3 |
| **E2** | Churn and hotspot persistence | Home building, multi-session | BSSID survival within hour vs across days | C3, C6 |
| **E3** | AP-scarcity curve to zero | Public datasets, subsampled | Room accuracy vs AP count, per model | C3 |
| **E4** | OCR model comparison | 20 nameplates, 6 conditions | Accuracy by condition, size, latency | C4 |
| **E5** | Fused recognition | Doorway photos, enrolment sets | Top-1, confident-wrong rate, by condition | C4 |
| **E6** | Motion drift per segment | Walk logs vs ground truth | Position error vs distance since fix | C1, C2 |
| **E7** | Prompting policy | Walkthrough logs | Prompts per journey vs wrong-room rate; Wi-Fi ablation | C2 |
| **E8** | Navigation trial | Participants new to the building | Arrival success, wrong turns, time, extra distance, prompts, NASA-TLX | C1 |
| **E9** | Off-route detection | Trial logs with deliberate wrong turns | Detection delay (steps), false alarms | C1 |

### E8 · The navigation trial, designed properly

- **Conditions:** shortest path, simplest path (fewest turns), risk-aware path. Same app, same prompts logic, only the planner changes.
- **Participants:** first-year students or visitors unfamiliar with the Main Block. Aim for at least 18–24 so conditions can be balanced; confirm the number with your guide.
- **Learning effect:** people learn the building as they walk. Give each participant a different destination per condition and rotate condition order (Latin square), or use a between-subjects design. Choose destination pairs where the three planners actually produce different routes — otherwise the comparison is empty.
- **Ground truth:** a team member follows at a distance and records wrong turns and arrival; the app logs time, distance and prompts.
- **Ethics:** written consent, no names in logs, guide and department sign-off before recruiting.
- **Analysis:** report effect sizes with confidence intervals, not just p-values; with this sample size, be modest.

# 08 Phased plan

Read the dependencies before the durations. Risk models and prompting need walk logs; walk logs need the app; the app needs the graph; the graph needs the floor walk. The core path fits in roughly three months; the extended phase runs after it.

| Phase | What | Depends on | Output |
|---|---|---|---|
| **Done** | Lab AP survey, throttled and unthrottled | — | Initial AP and hotspot counts (to be reconciled) |
| **P1 · wk 1–2** | Guide permission. Floor walk, sketch, nameplate and doorway photos. Anonymised scan logger. Churn collection begins. Send logger to other colleges. | — | rooms.csv, edges.csv, sign photos, logger |
| **P2 · wk 2–5** | Working app: ML Kit OCR, fuzzy matching, SQLite graph, Dijkstra, step and heading tracking snapped to graph. | P1 | Demoable app |

| Phase | What | Depends on | Output |
|---|---|---|---|
| **P3 · wk 3–6** | Offline, in parallel: E1, E2, E3, E4. Literature review paper drafting. | P1 | Regime results, DL table |
| **P4 · wk 5–8** | Fused recognition (E5). Pilot walks with logging. Motion drift (E6). First risk model. | P2 | Recognition result, risk labels |
| **P5 · wk 7–10** | Risk-aware planner, λ sweep, avoid-or-ask logic, prompting policy (E7), off-route detector. | P4 | Navigation method |
| **P6 · wk 10–12** | Navigation trial (E8), off-route evaluation (E9). Dataset packaged with datasheet; anonymisation verified. | P5, approvals | Headline result, dataset |
| **P7 · extended** | On-device fine-tuned OCR, neural corrector, closed-loop risk updates, conformal prediction, AR arrow. | Working system | Upgrades |
| **P8 · wk 11–14** | Figures, technical paper, arXiv preprint, guide review, submission. | P6 | Submitted papers |

**THE THINGS THAT WOULD SINK THIS**

1. **Delaying the floor walk.** Everything downstream waits on rooms.csv and edges.csv.
2. **Late approval for the navigation trial.** The headline claim depends on it. Ask for sign-off in week 1, not week 9.
3. **Building models before the app works.** A clever risk model attached to nothing does not pass a demo.
4. **Unopened references.** Every citation in Section 10 must be opened and the attributed claim confirmed before submission.
5. **Unmeasured claims.** “Many Indian colleges have few APs” and every “no study has…” need evidence — E1 and a documented literature search.

# 09 Explaining it — to the guide, and in the viva

## The core pitch, memorised by all four

> *Our college has only one or two Wi-Fi access points. Almost every network your phone sees is someone's hotspot, and it disappears after class — and [once E1 confirms it] we found the same pattern in other colleges. Research indoor navigation assumes hundreds of stable access points, and apps just send you along the shortest path. Here that path goes past faded signs and confusing junctions where people get lost. So our app recognises rooms from the door itself, tracks your steps between doors, and plans the route it can actually guide you along — asking you to check a sign only where a wrong turn would cost you.*

## Paper titles

> **Technical paper:** Risk-Aware Indoor Navigation for Infrastructure-Poor Buildings: Planning Routes a Smartphone Can Reliably Guide
>
> **Literature review paper:** Indoor Navigation Beyond Dense Infrastructure: A Review of Localisation, Sign Reading and Route Planning in Resource-Constrained Buildings
>
> **System name in the paper:** RAIN — e.g. “We present RAIN, a risk-aware indoor navigation system…” Search the name on Google Scholar, GitHub and the Play Store before submission to rule out clashes.
>
> Avoid “first,” “novel” or “accurate” in any title. If the navigation trial shows shortest paths were already safe, keep this title and report that result honestly.

## Contribution statement for the paper

*We present a phone-only indoor navigation system for infrastructure-poor academic buildings with zero to two access points and degraded signage. We characterise this regime empirically across multiple colleges and show how radio fingerprinting collapses as access points approach zero. We recognise rooms by fusing on-device text recognition with doorway appearance, and track users between recognitions with motion sensors over a topological floor graph. Our central contribution is risk-aware route planning: a learned model predicts where users are likely to go wrong, and the same model drives both route selection and an uncertainty-aware policy that decides when to prompt the user to scan a sign, trading distance and user effort against navigation error. A participant study compares shortest, simplest and risk-aware routes. We release an anonymised benchmark of degraded signage, labelled RF scans, motion traces and trial logs.*

## “Where is the AI, and did it actually help?”

Machine learning makes the navigation decisions — which route to take, when to ask, when the user has gone wrong — because those are small-data prediction problems. Deep learning handles perception — reading degraded signs and recognising doorways — because that is our primary sensor. Where learning did not help, such as Wi-Fi fingerprinting with one or two access points, we show it with a controlled experiment and keep the simpler method.

## Hard questions

| Question | Answer |
|---|---|
| **“This is just Google Maps.”** | Google needs venue-supplied floor plans and routes by distance. We measure a regime with no plans, almost no Wi-Fi, and route by guidability. |
| **“Isn't this just simplest-path routing?”** | Simplest path uses fixed rules about turns. Our risk is learned from measured sign reliability and drift, and it also decides when to prompt. Simplest path is one of our baselines. |
| **“The building is small — why not shortest path?”** | Maybe it is enough. The trial measures that directly, and the λ curve shows exactly how much reliability each extra metre buys. If shortest path is already safe, we report it. |
| **“OCR is old technology.”** | Yes. What is new is reliability on signage nobody has tested, fused with appearance when signs fail, and feeding a routing decision. |
| **“Only one building.”** | The regime is measured across several colleges; the scarcity curve comes from public datasets. The navigation trial is one building, and we say so. |
| **“Why not just install more Wi-Fi?”** | That is the point. Nobody will install access points for a navigation app. Our design assumes they will not. |
| **“Your model didn't beat the baseline.”** | Here is where it did and where it did not, stratified by condition. Negative results are reported as results. |

### Rules for the room

| Do | Do not |
|---|---|
| Lead with the problem: one or two APs, unreadable signs, people getting lost | Claim to be first to read door signs, filter hotspots, or route for simplicity |
| Say what you are not claiming, unprompted | Argue the Phase I idea was novel — it was not |
| Show the λ trade-off curve and the trial result together | Promise accuracy or success figures you have not measured |
| Report negative results as results | List features to sound impressive — feature count is not novelty |
| Frame deferred work as scoping discipline | Cite a paper nobody on the team has opened |

# 10 References to open and verify

Every item below must be opened by a team member and the attributed claim confirmed in the source text before it appears in any submission. Items marked **[NEW]** were added in v4 and their bibliographic details have **not** yet been verified — check authors, year, venue and DOI against the publisher record. Vendor documentation may be cited only for engineering facts.

**Navigation-side prior art — read before writing Claim 1 [NEW]**

- Duckham & Kulik. “Simplest” paths: automated route selection for navigation. COSIT 2003 (LNCS). [NEW]
- Grum. Danger of getting lost: optimize a path to minimize risk. CORP 2005. [NEW]
- Vanclooster et al. Empirical evaluation of the least risk path algorithm for indoor wayfinding. [NEW — confirm title, journal and year]
- Search task: wrong-way / off-route detection in pedestrian navigation, and learned route-choice or route-difficulty models. [NEW — no source located yet]

**Motion tracking and place recognition [NEW]**

- Wang et al. No Need to War-Drive: Unsupervised Indoor Localization (UnLoc). MobiSys 2012 — landmark-reset dead reckoning. [NEW]
- Arandjelović et al. NetVLAD: CNN architecture for weakly supervised place recognition. CVPR 2016. [NEW]
- Google MediaPipe Image Embedder documentation — engineering facts only. [NEW]

**Already in related work (from v3)**

- Bolliger. Redpin. MELT '08. DOI 10.1145/1410012.1410025
- Jung et al. QRLoc. IEEE Xplore document 6530571
- Kim & Suh. Heimdall (2024). LLM-Loc (2026). TextInPlace — Tao et al., IROS 2025, arXiv 2503.06501
- Tang et al. Static vs. Dynamic Databases for Indoor Localization based on WiFi Fingerprinting. arXiv 2402.12756 (XJTLU; hotspot contamination)

**Sign reading and OCR**

- Sadeghi, Valaee & Shirani. OCRAPOSE. ICASSP 2015, DOI 10.1109/ICASSP.2015.7178215. OCRAPOSE II, arXiv 1704.05591
- Evaluating OCR Performance for Assistive Technology. arXiv 2602.02223; DOI 10.1080/17483107.2026.2673178
- Guided Text Spotting for assistive indoor navigation (ISVC)
- PP-OCR. arXiv 2009.09941; PP-OCRv5 model card. Google ML Kit Text Recognition v2 documentation (engineering limits only)

**Sparse-AP evidence**

- Ye et al. EdgeLoc. arXiv 2009.05780 (curve stops at 20 APs)
- Bhatia et al. Locaris. arXiv 2510.11926
- Narasimman & Alphones. DumbLoc. IEEE Sensors Journal 24(9), 2024. DOI 10.1109/JSEN.2024.3374415
- Kumar, Torres-Sospedra & Chaurasiya. Single-AP WiFi fingerprinting datasets. CEUR-WS Vol-3434 paper 6

**Confidence, prompting and evaluation**

- Conformal Prediction for Indoor Positioning with Correctness Coverage Guarantees. arXiv 2505.01810
- Feng, Nguyen & Luo. Sensors 2026, DOI 10.3390/s26010284
- Bıyık et al. Asking Easy Questions. arXiv 1910.04365
- Hart & Staveland. Development of NASA-TLX. 1988. [NEW]
- Gebru et al. Datasheets for Datasets. Communications of the ACM, 2021. [NEW]

**Deployment constraints**

- Nowicki & Skrzypczyński. Sensors 2019, DOI 10.3390/s19173657 — scan throttling
- Jaisinghani et al. arXiv 2101.09515 (ACM TCPS 2021) — scan reduction and cardinality mismatch

## Venues

arXiv preprint first. Then IEEE INDICON, IEEE INDISCON (Graduate Research Forum), or a Scopus-indexed Springer conference. With the navigation trial, an HCI or ubiquitous-computing workshop is also worth checking. Not IEEE Access for a single-floor study. Verify every venue: search its name with “predatory,” check that researchers you recognise have presented there, and ask the department for a safe-list.

*Action Plan v4. Supersedes Action Plan v3; the team is unchanged and the project is renamed from SHINS 2.0 to RAIN (Risk-Aware Indoor Navigation). RF counts from the single-lab survey must be reconciled into physical APs and BSSIDs and confirmed across rooms and colleges before appearing in a paper. The scan-throttling observation must be re-tested against cached-result behaviour. Model sizes and latencies must be re-measured on the team's phones. References marked [NEW] are unverified. Hotspot identifiers must be anonymised at capture. The navigation trial requires guide and department approval and participant consent before recruitment.*